



College of
EDUCATION

PATH ANALYSIS

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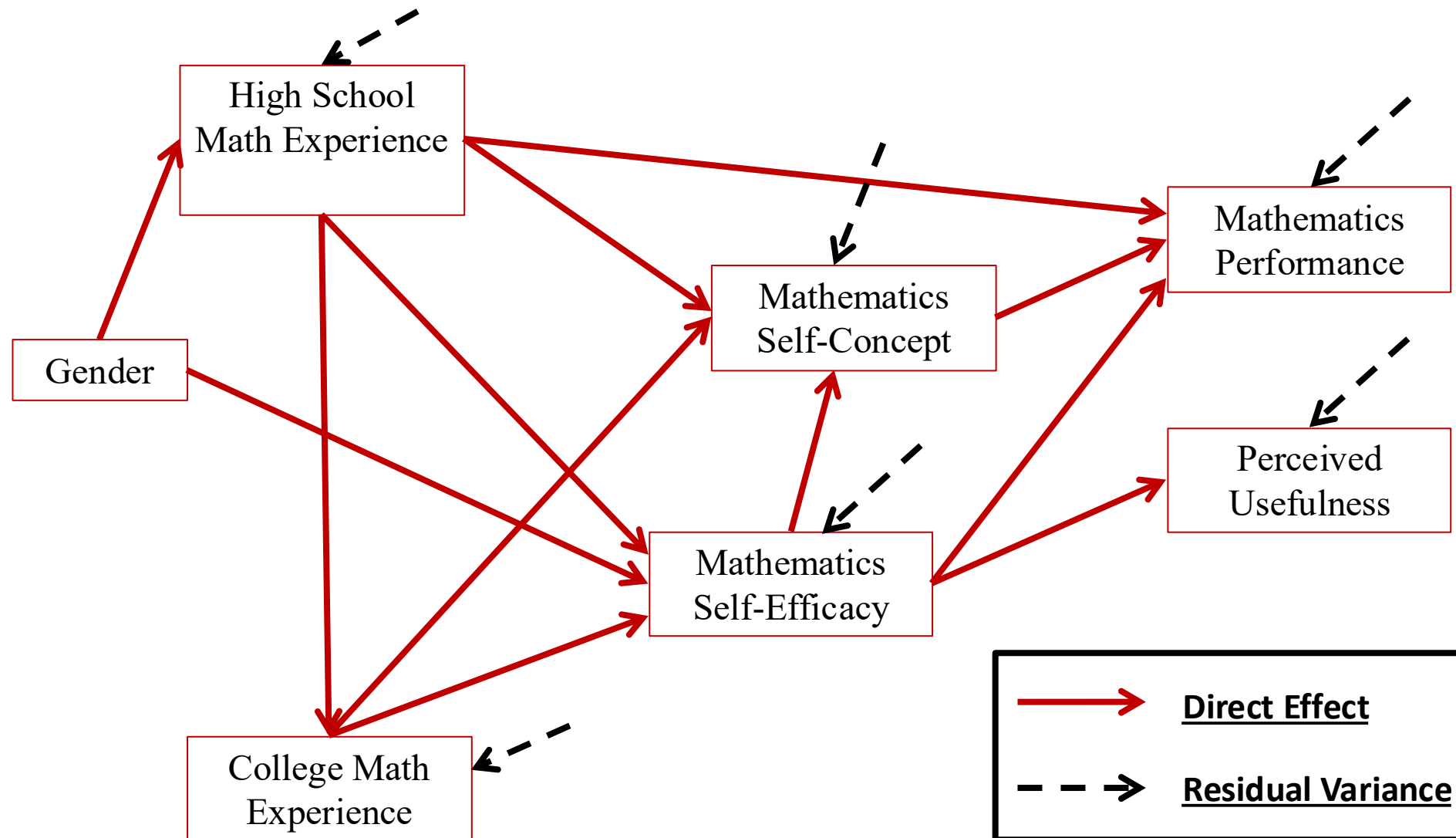
- Identifying path models analysis
 - Some basics to ensure the model is estimable

- Path analysis details:
 - Model modification
 - Direct and indirect effects

- Additional issues in path analysis
 - Variable considerations

- Data are simulated based on the results reported in:
Pajares, F., & Miller, M. D. (1994). Role of self-efficacy and self-concept beliefs in mathematical problem solving: a path analysis. *Journal of Educational Psychology*, 86, 193-203.
- Sample of 350 undergraduates (229 women, 121 men)
 - In simulation, 10% of variables were missing (using missing completely at random mechanism)
- Note: simulated data characteristics differ from actual data (some variables extend beyond their official range)
 - Simulated using Multivariate Normal Distribution
 - Some variables had boundaries that simulated data exceeded
 - Results will not match exactly due to missing data and boundaries

- Gender (1 = male; 0 = female)
- Math Self-Efficacy (MSE)
 - Reported reliability of .91
 - Assesses math confidence of college students
- Perceived Usefulness of Mathematics (USE)
 - Reported reliability of .93
- Math Anxiety (MAS)
 - Reported reliability ranging from .86 to .90
- Math Self-Concept (MSC)
 - Reported reliability of .93 to .95
- Prior Experience at High School Level (HSL)
 - Self report of number of years of high school during which students took mathematics courses
- Prior Experience at College Level (CC)
 - Self report of courses taken at college level
- Math Performance (PERF)
 - Reported reliability of .788
 - 18-item multiple choice instrument (total of correct responses)



IDENTIFICATION OF PATH MODELS

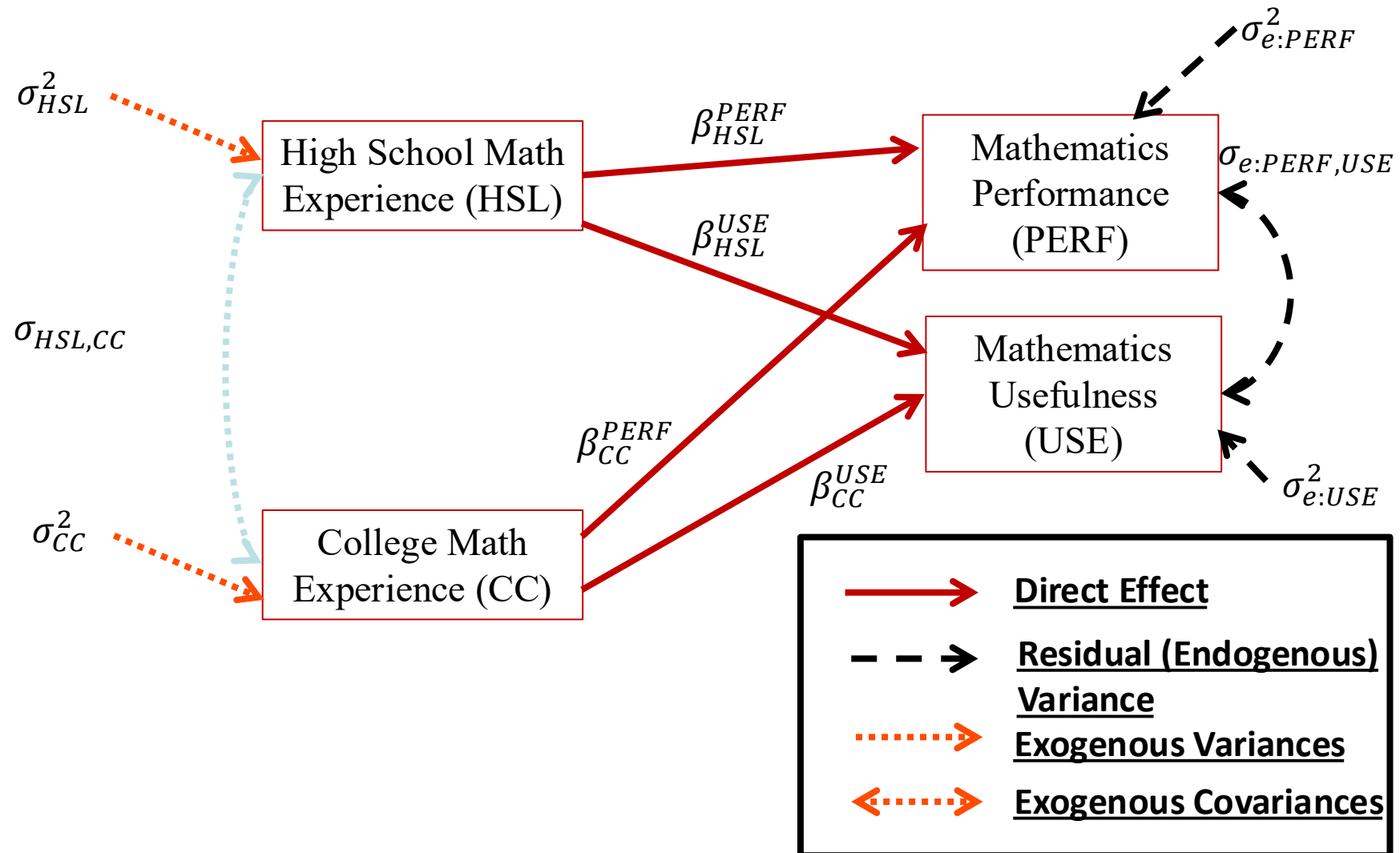
- Model identification is necessary for statistical models to have meaningful results
 - Cannot have too many unknown values (parameters) and not enough “places to put” the parameters in the model distribution (the multivariate normal distribution)
- For path models, identification can be a very difficult thing to understand
- Because of their unique structure, path models must have identification in two ways:
 - “Globally” – so that the total number of parameters does not exceed the total number of means, variances, and covariances of the endogenous and exogenous variables
 - “Locally” – so that each individual equation is identified
- Identification is guaranteed if a model is both “globally” and “locally” identified

- A necessary but not sufficient condition for path models is that of having equal to or fewer model parameters than there are parameters in the joint distribution implied by the model
- As the path models we discuss today assume the multivariate normal distribution, we have two matrices of parameters with which to work:
 - The elements of the mean vector and
 - The unique elements of the covariance matrix
- For the MVN, the so-called T-rule states that a model must have equal to or fewer parameters than the unique elements of the covariance matrix of all endogenous and exogenous variables
 - Let $s = p + q$, the total of all endogenous (p) and exogenous (q) variables
 - Then the total unique elements are $\frac{s(s+1)}{2}$

- The classical definition of the “T-rule” counts the following entities as model parameters:
 - Direct effects (regression slopes)
 - Residual variances
 - Residual covariances
 - Exogenous variances
 - Exogenous covariances

- Missing from this list are:
 - The set of exogenous variable means
 - The set of intercepts for endogenous variables

- Each of the missing entities are part of the lavaan likelihood function, but are considered “saturated” so no additional parameters can be added
 - These do not enter into the equation for the covariance matrix of the endogenous and exogenous variables



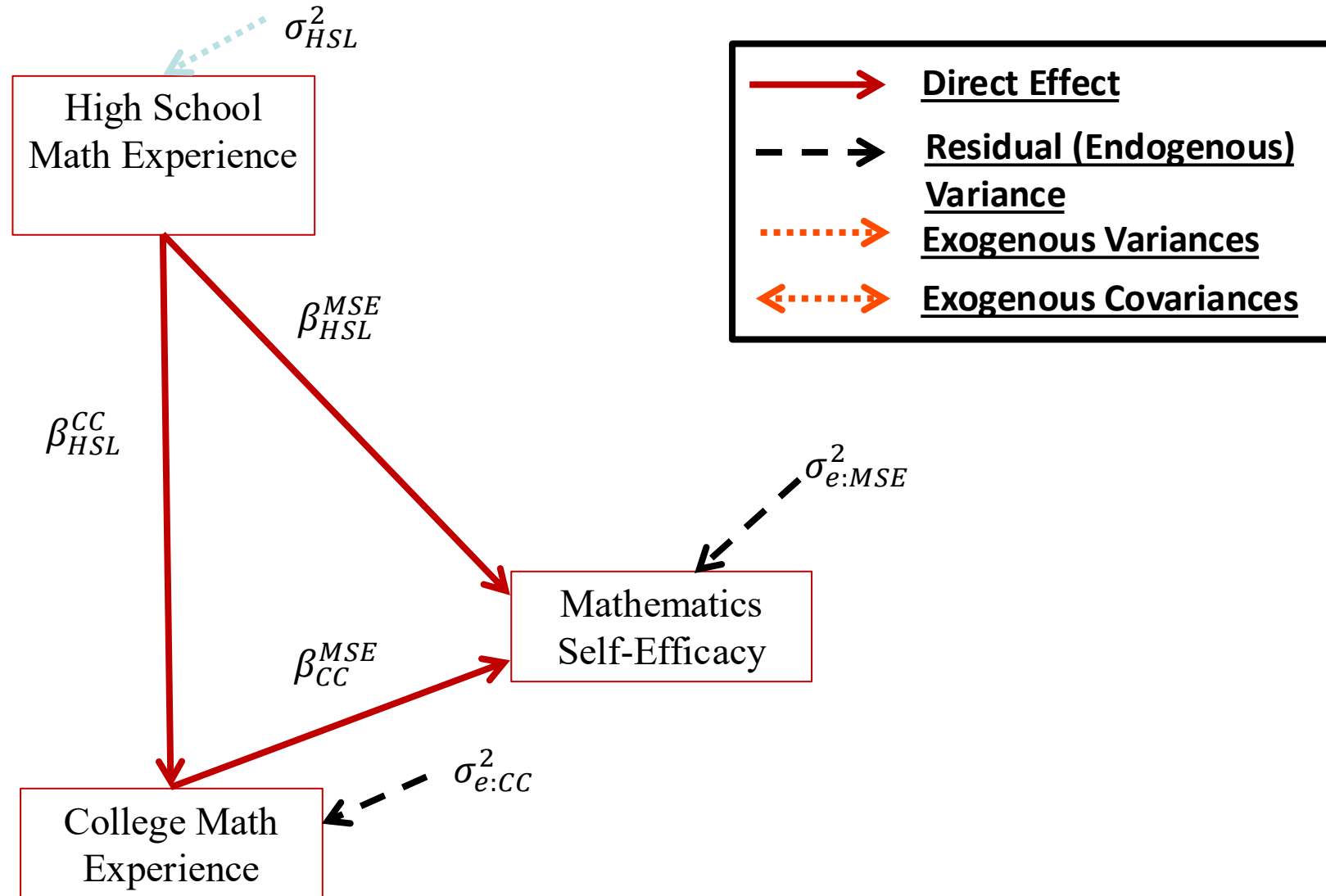
Model	Endogenous Variables (p)	Exogenous Variables (q)	Unique Covariance Matrix Elements	Model Parameters (excluding Exo. Var. means)	Identification Status
Multivariate Regression Full Model	2	2	$\frac{4*(4+1)}{2} = 10$	10: $\sigma_{HSL}^2, \sigma_{HSL,CC}, \sigma_{CC}^2, \beta_{HSL}^{PERF}, \beta_{CC}^{PERF}, \beta_{HSL}^{USE}, \beta_{CC}^{USE}, \sigma_{e:PERF}^2, \sigma_{e:USE}^2, \sigma_{e:PERF,USE}$	Just Identified

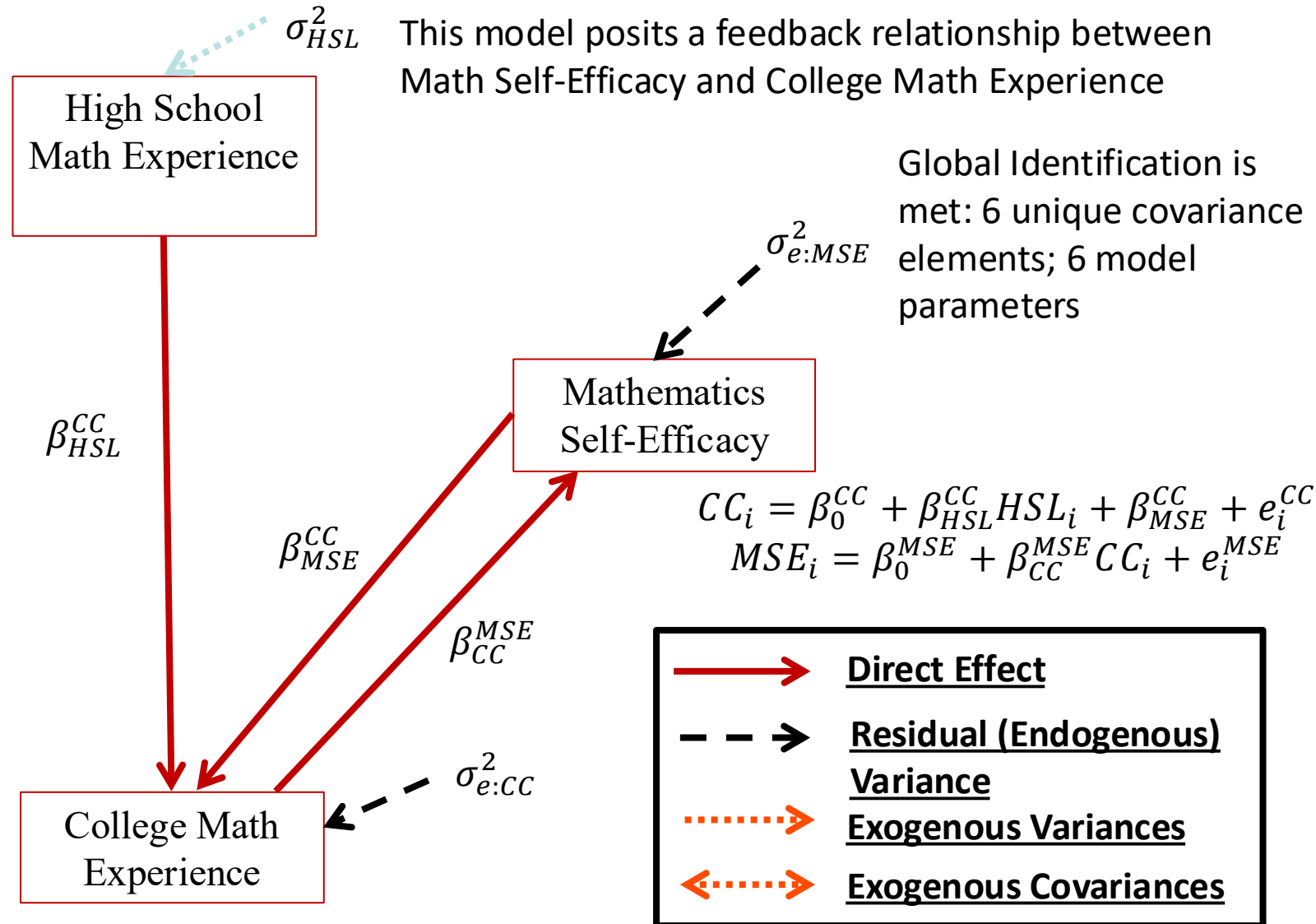
- **Just-Identified:**
number of covariances == number of model parameters
 - Necessary for identification, but no model fit indices available

- **Over-Identified:**
number of covariances > number of model parameters
 - Necessary for identification; model fit indices available

- **Under-Identified:**
number of covariances < number of model parameters
 - **Model is NOT IDENTIFIED:** No results available

- For most research designs, global identification will suffice
 - For the most part, **recursive path models** will be identified if the “t-rule” is met
- A **recursive path model** is one where the direct effects are unidirectional – no feedback loops
 - Our path model is an example of a recursive path model
- A **non-recursive path model** is one where the direct effects are bidirectional for some variables – feedback loops are present
 - Difficult to envision using cross-sectional data
 - More frequent in econometrics
 - Different estimation algorithms used (see the next few slides)

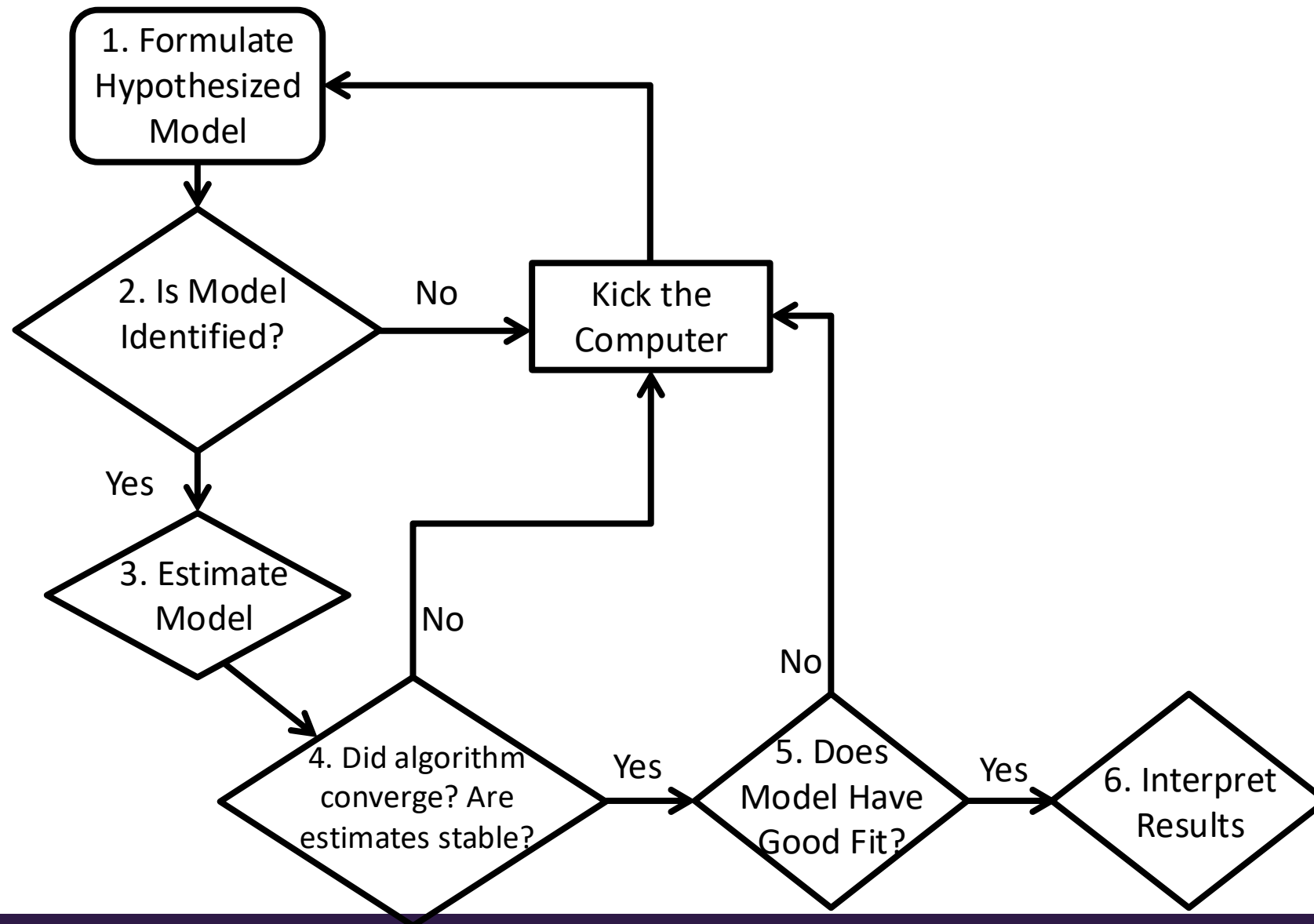


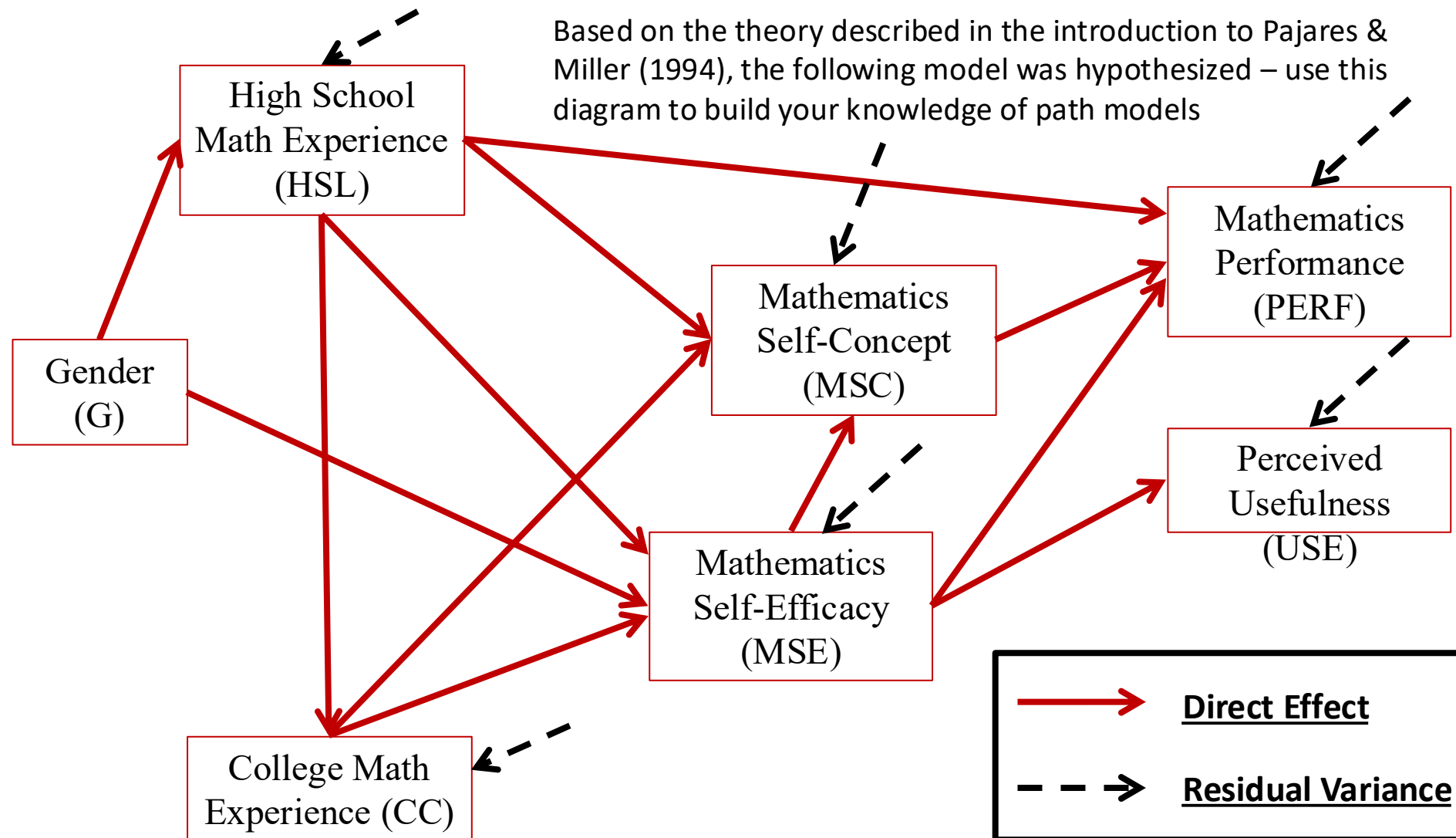


- If you have a recursive model (no feedback loops) make sure:
 - # of model parameters $\leq$ # of unique covariance elements
 - No undirected paths (residual covariance) connecting variables with direct effects
 - Does not make sense to say one variable causes another yet their correlation is unexplained

- If you have a non-recursive model (feedback loops):
 - Think critically about whether such a model can be investigated by your data
 - Cross-sectional data may need a longitudinal design
 - Attempt to determine if the model meets the rank condition
 - Investigate model output for irregularities (very large effects relative to the scale of the variables)

THE FINAL PATH MODEL: PUTTING IT ALL TOGETHER





- How many variables are in our model? 7
 - Gender, HSL, CC, MSC, MSE, PERF, and USE
- How many variables are endogenous? 6
 - HSL, CC, MSC, MSE, PERF, and USE
- How many variables are exogenous? 1
 - Gender
- Is the model recursive or non-recursive?
 - Recursive – no feedback loops present

➤ Is the model identified?

- Check the t-rule first (and only as it is recursive)
- How many covariance terms are there in the all-variable matrix?

$$\frac{7*(7 + 1)}{2} = 28$$

➤ How many model parameters are to be estimated?

- 12 direct paths
- 6 residual variances
- 1 variance of the exogenous variable
- **(19 model parameters for the covariance matrix)**
- 6 endogenous variable intercepts
 - Not relevant for t-rule identification, but counted in Mplus

➤ The model is over-identified

- 28 total variance/covariances but 19 model parameters
- We can use lavaan to run our analysis

- The path model from can be re-expressed in the following 6 endogenous variable regression equations:

$$1. \quad HSL_i = \beta_0^{HSL} + \beta_G^{HSL} G_i + e_i^{HSL}$$

$$2. \quad CC_i = \beta_0^{CC} + \beta_{HSL}^{CC} HSL_i + e_i^{CC}$$

$$3. \quad MSE_i = \beta_0^{MSE} + \beta_G^{MSE} G_i + \beta_{HSL}^{MSE} HSL_i + \beta_{CC}^{MSE} CC_i + e_i^{MSE}$$

$$4. \quad MSC_i = \beta_0^{MSC} + \beta_{HSL}^{MSC} HSL_i + \beta_{CC}^{MSC} CC_i + \beta_{MSE}^{MSC} MSE_i + e_i^{MSC}$$

$$5. \quad USE_i = \beta_0^{USE} + \beta_{MSE}^{USE} MSE_i + e_i^{USE}$$

$$6. \quad PERF_i = \beta_0^{PERF} + \beta_{HSL}^{PERF} HSL_i + \beta_{MSE}^{PERF} MSE_i + \beta_{MSC}^{PERF} MSC_i + e_i^{PERF}$$

- Having (1) constructed our model and (2) verified it was identified using the t-rule and that it is a recursive model, the next step is to (3) estimate the model with lavaan

```
model06.syntax = "  
#endogenous variable equations  
perf ~ hsl + msc + mse  
use ~ mse  
mse ~ hsl + cc + gender  
msc ~ mse + cc + hsl  
cc ~ hsl  
hsl ~ gender  
  
#endogenous variable intercepts  
perf ~ 1  
use ~ 1  
mse ~ 1  
msc ~ 1  
cc ~ 1  
hsl ~ 1  
  
#endogenous variable residual variances  
perf ~~ perf  
use ~~ use  
mse ~~ mse  
msc ~~ msc  
cc ~~ cc  
hsl ~~ hsl  
  
#endogenous variable residual covariances  
#none specified in the original model so these have zeros:  
perf ~~ 0*use + 0*mse + 0*msc + 0*cc + 0*hsl  
use ~~ 0*mse + 0*msc + 0*cc + 0*hsl  
mse ~~ 0*msc + 0*cc + 0*hsl  
msc ~~ 0*cc + 0*hsl  
cc ~~ 0*hsl  
  
#exogeneous variables put into likelihood function:  
  
#means(intercepts)  
gender ~ 1  
  
#variances  
gender ~~ gender  
  
"  
|
```

- First, we check convergence:

```
> summary(model06.fit, fit.measures=TRUE, standardized=TRUE)  
lavaan 0.6-19 ended normally after 68 iterations
```

- lavaans's algorithm converged
- Second, we check for abnormally large standard errors
 - None too big, relative to the size of the parameter
 - Indicates identified model
- Third, we look at the model fit statistics:

Model Test User Model:		
	Standard	Scaled
Test Statistic	58.896	58.913
Degrees of freedom	9	9
P-value (Chi-square)	0.000	0.000
Scaling correction factor		1.000
Yuan-Bentler correction (Mplus variant)		

This is a likelihood ratio (deviance) test comparing our model (H_0) with the saturated model – The saturated model fits much better (but that is typical).

Model Test Baseline Model:		
	Standard	Scaled
Test statistic	619.926	629.882
Degrees of freedom	21	21
P-value	0.000	0.000
Scaling correction factor		0.984

This compares the independence model (H_0) to the saturated model (H_1) – it indicates that there is significant covariance between variables

User Model versus Baseline Model:		
	Standard	Scaled
Comparative Fit Index (CFI)	0.917	0.918
Tucker-Lewis Index (TLI)	0.806	0.809
Robust Comparative Fit Index (CFI)		0.918
Robust Tucker-Lewis Index (TLI)		0.809

The CFI estimate is .918 and the TLI is .809. Good fit is considered 0.95 or higher.

Loglikelihood and Information Criteria:		
	Standard	Scaled
Loglikelihood user model (H_0)	-6126.013	-6126.013
Scaling correction factor for the MLR correction		0.937
Loglikelihood unrestricted model (H_1)	-6096.565	-6096.565
Scaling correction factor for the MLR correction		0.953
Akaike (AIC)	12304.025	12304.025
Bayesian (BIC)	12404.332	12404.332
Sample-size adjusted Bayesian (SABIC)	12321.850	12321.850

The RMSEA estimate is 0.126. Good fit is considered 0.05 or less.

Root Mean Square Error of Approximation:		
	Standard	Scaled
RMSEA	0.126	0.126
90 Percent confidence interval - lower	0.096	0.096
90 Percent confidence interval - upper	0.157	0.157
P-value H_0 : RMSEA $\leq$ 0.050	0.000	0.000
P-value H_0 : RMSEA $\geq$ 0.080	0.994	0.994
Robust RMSEA		0.140
90 Percent confidence interval - lower		0.108
90 Percent confidence interval - upper		0.175
P-value H_0 : Robust RMSEA $\leq$ 0.050		0.000
P-value H_0 : Robust RMSEA $\geq$ 0.080		0.999

The average standardized residual covariance is 0.056. Good fit is less than 0.05.

Standardized Root Mean Square Residual:		
	Standard	Scaled
SRMR	0.056	0.056

Based on the model fit statistics, we can conclude that our model **does not** do a good job of approximating the covariance matrix – so we cannot make inferences with these results (biased standard errors and effects may occur)

- Now that we have concluded that our model fit is poor we must modify the model to make the fit better
 - Our modifications are purely statistical – which draws into question their generalizability beyond this sample
- **Model modification should be guided by theory**
 - However, we can inspect the normalized residual covariance matrix (like z-scores) to see where our biggest misfit occurs

```
> residuals(model06.fit, type="normalized")
$type
[1] "normalized"

$cov
      perf    use    mse    msc    cc    hsl gender
perf -0.076
use  -0.159  0.041
mse  -0.071 -0.110 -0.086
msc   0.059  5.051 -0.039  0.043
cc   -0.028  0.720 -0.377 -0.161  0.046
hsl   0.006  0.559  0.085  0.105 -0.034  0.039
gender -1.522 -0.027 -0.422 -1.452 -2.567  0.091  0.000

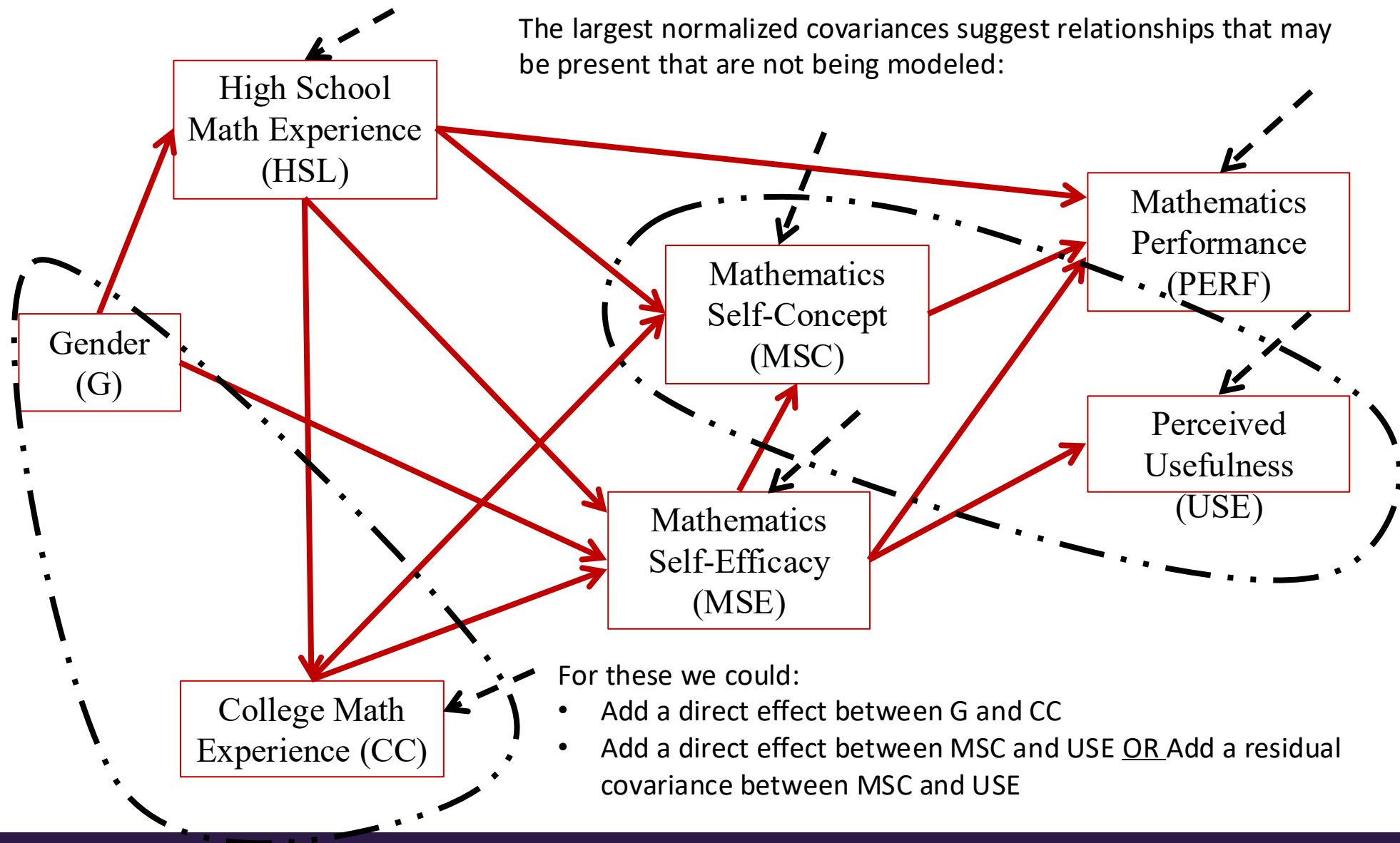
$mean
      perf    use    mse    msc    cc    hsl gender
-0.014  0.126  0.012  0.211  0.009  0.004  0.000
```

Two normalized residual covariances are bigger than +/-1.96:
MSC with USE and
CC with Gender

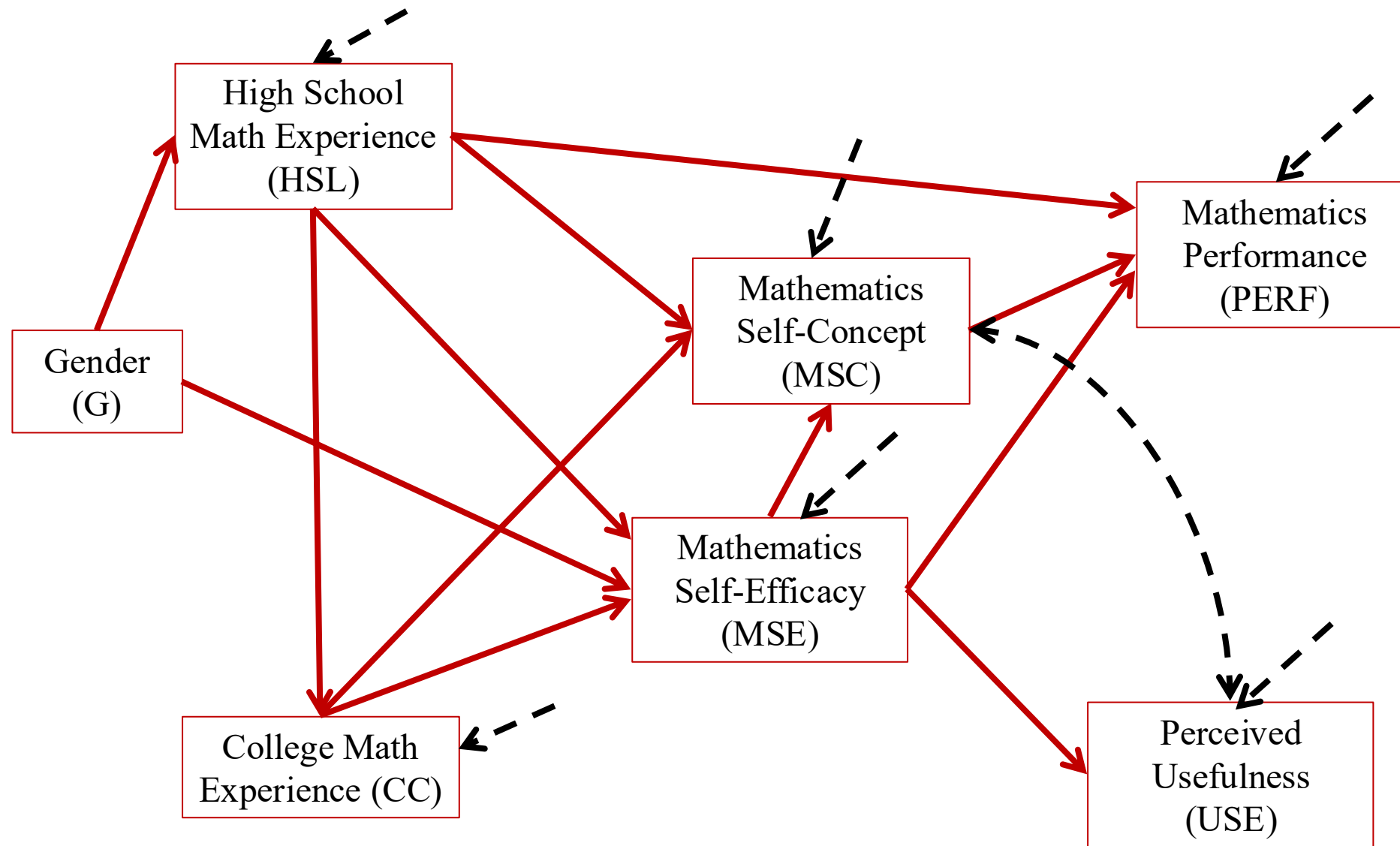
- As we used Maximum Likelihood to estimate our model, another useful feature is that of the modification indices
- Modification indices (also called Score or LaGrangian Multiplier) tests that attempt to suggest the change in the log-likelihood for adding a given model parameter (larger values indicate a better fit for adding the parameter)

```
> modindices(model06.fit)
```

	lhs	op	rhs	mi	epc	sepc.lv	sepc.all	sepc.nox
31	use	~~	msc	41.517	70.912	70.912	0.386	0.386
52	use	~	msc	40.032	0.451	0.451	0.490	0.490
60	msc	~	use	41.517	0.299	0.299	0.275	0.275



- The modification indices have three large values:
 - A direct effect predicting MSC from USE
 - A direct effect predicting USE from MSC
 - A residual covariance between USE and MSC
- Note: the MI value is -2 times the change in the log-likelihood and the EPC is the expected parameter value
 - The MI is like a 1 DF Chi-Square Deviance test
 - Values greater than 3.84 are likely to be significant changes in the log-likelihood
- Because all three happen for the same variable, we can only choose one
 - This is where theory would help us decide
- As we do not know theory, we will choose to add a residual covariance between USE and MSC
 - Their covariance is **unexplained** by the model – not a great theoretical statement (but will allow us to make inferences if the model fits)
 - MI = 41.529
 - EPC = 70.912



- Now we must start over with our path model decision tree
 - The model is identified (now 20 parameters < 28 covariances)
 - lavaan estimation converged; Standard errors look acceptable
- Model fit statistics:

Estimator	ML	Robust
Minimum Function Test Statistic	14.827	14.393
Degrees of freedom	8	8
P-value (Chi-square)	0.063	0.072
Scaling correction factor for the Yuan-Bentler correction (Mplus variant)		1.030

The comparison with the saturated model suggests our model fits statistically

Root Mean Square Error of Approximation:

RMSEA		0.049	0.048
90 Percent Confidence Interval	0.000	0.088	0.000 0.086
P-value RMSEA <= 0.05		0.457	0.484

The RMSEA is 0.048, which indicates good fit

User model versus baseline model:

Comparative Fit Index (CFI)	0.989	0.990
Tucker-Lewis Index (TLI)	0.970	0.972

The CFI and TLI both indicate good fit

Standardized Root Mean Square Residual:

SRMR	0.035	0.035
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The SRMR also indicates good fit

Therefore, we can conclude the model adequately approximates the covariance matrix – meaning we can now inspect our model parameters...

Model Parameter Investigation

	Estimate	Std.err	Z-value	P(> z)	Std.lv	Std.all
Regressions:						
perf ~						
hsl	0.153	0.107	1.432	0.152	0.153	0.068
msc	0.037	0.009	4.147	0.000	0.037	0.215
mse	0.139	0.013	10.700	0.000	0.139	0.557
use ~						
mse	0.277	0.073	3.803	0.000	0.277	0.209
mse ~						
hsl	4.138	0.406	10.203	0.000	4.138	0.459
cc	0.393	0.105	3.723	0.000	0.393	0.194
gender	4.168	1.160	3.593	0.000	4.168	0.166
msc ~						
mse	0.736	0.066	11.119	0.000	0.736	0.512
cc	0.519	0.117	4.434	0.000	0.519	0.179
hsl	2.824	0.593	4.764	0.000	2.824	0.218
cc ~						
hsl	0.662	0.247	2.686	0.007	0.662	0.149
hsl ~						
gender	0.208	0.154	1.348	0.178	0.208	0.075
Covariances:						
use ~						
msc	70.249	10.358	6.782	0.000	70.249	0.380

Some parameters are not significant...we could omit those, but we'll leave them for now

Intercepts:

perf	1.023	0.782	1.309	0.191	1.023	0.344
use	32.162	5.393	5.964	0.000	32.162	2.035
mse	47.738	2.237	21.339	0.000	47.738	4.006
msc	-23.369	4.484	-5.211	0.000	-23.369	-1.364
cc	7.072	1.268	5.576	0.000	7.072	1.201
hsl	4.843	0.091	53.390	0.000	4.843	3.663
gender	0.346	0.025	13.599	0.000	0.346	0.727

Variances:

perf	3.763	0.309	12.173	0.000	3.763	0.427
use	238.854	19.097	12.507	0.000	238.854	0.956
mse	97.294	7.758	12.541	0.000	97.294	0.685
msc	142.912	10.793	13.241	0.000	142.912	0.487
cc	33.923	2.456	13.813	0.000	33.923	0.978
hsl	1.738	0.126	13.813	0.000	1.738	0.994
gender	0.226	0.008	28.835	0.000	0.226	1.000

- We can interpret the std.all standardized parameter estimates for all variables except gender
 - It is not continuous, so SD of gender does not make sense
- A 1-SD increase in HSL means CC increases by 0.149 SD

	Estimate	Std.err	Z-value	P(> z)	Std.lv	Std.all
Regressions:						
perf ~						
hsl	0.153	0.107	1.432	0.152	0.153	0.068
msc	0.037	0.009	4.147	0.000	0.037	0.215
mse	0.139	0.013	10.700	0.000	0.139	0.557
use ~						
mse	0.277	0.073	3.803	0.000	0.277	0.209
mse ~						
hsl	4.138	0.406	10.203	0.000	4.138	0.459
cc	0.393	0.105	3.723	0.000	0.393	0.194
gender	4.168	1.160	3.593	0.000	4.168	0.166
msc ~						
mse	0.736	0.066	11.119	0.000	0.736	0.512
cc	0.519	0.117	4.434	0.000	0.519	0.179
hsl	2.824	0.593	4.764	0.000	2.824	0.218
cc ~						
hsl	0.662	0.247	2.686	0.007	0.662	0.149
hsl ~						
gender	0.208	0.154	1.348	0.178	0.208	0.075
Covariances:						
use ~						
msc	70.249	10.358	6.782	0.000	70.249	0.380
mse						

- The STDY standardization does not standardize by the SD of the X variable
 - So its interpretation makes sense for Gender (1 = male)

Here, males have an average MSE (intercept) that is .166 SD higher than females

```
> standardizedSolution(model03.fit, type="std.nox")
```

	lhs	op	rhs	est.std	se	z	pvalue
1	perf	~	hs1	0.068	0.048	1.432	0.152
2	perf	~	msc	0.215	0.052	4.147	0.000
3	perf	~	mse	0.557	0.052	10.700	0.000
4	use	~	mse	0.209	0.055	3.803	0.000
5	mse	~	hs1	0.459	0.045	10.203	0.000
6	mse	~	cc	0.194	0.052	3.723	0.000
7	mse	~	gender	0.166	0.046	3.593	0.000
8	msc	~	mse	0.512	0.046	11.119	0.000
9	msc	~	cc	0.179	0.040	4.434	0.000
10	msc	~	hs1	0.218	0.046	4.764	0.000
11	cc	~	hs1	0.149	0.055	2.686	0.007
12	hs1	~	gender	0.075	0.055	1.348	0.178

- High School Experience is a significant predictor of College Experience
 - More High School Experience means more College Experience

- High School Experience, College Experience, and Gender are significant predictors of Math Self-Efficacy
 - More High School and College Experience means higher Math Self-Efficacy
 - Men have higher Math Self-Efficacy than Women

- High School Experience, College Experience, and Math Self-Efficacy are significant predictors of Math Self-Concept
 - More High School and College Experience and higher Math Self-Efficacy mean higher Math Self-Concept

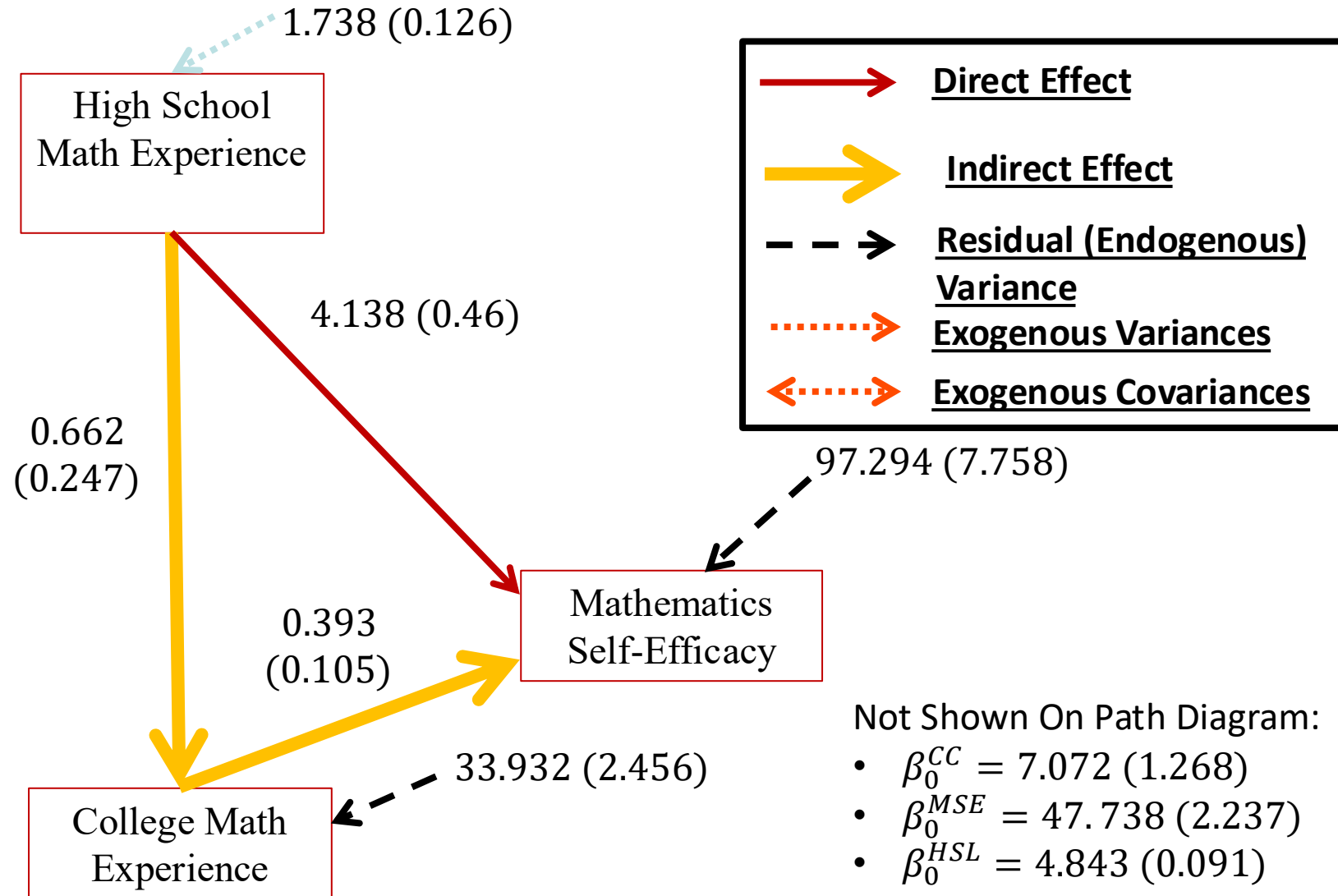
- Higher Math Self-Efficacy means significantly higher Perceived Usefulness
- Higher Math Self-Efficacy and Math Self-Concept result in higher Math Performance scores
 - High school experience was not significantly related to performance
- Math Self-Concept and Perceived Usefulness have a significant residual covariance

- The R^2 for each endogenous variable:

```
> inspect(model03.fit, what="r2")  
perf    use    mse    msc    cc    hs1  
0.573 0.044 0.315 0.513 0.022 0.006
```

- Note how high school experience, college experience, and perceived usefulness both have low percentages of variance accounted for by the model
 - We could have increased the R^2 for USE by adding the direct path between MSC and USE instead of the residual covariance

- Because High School Experience (HSL) predicted College Experience (CC) and College Experience (CC) predicted Math Self-Efficacy (MSE), an indirect path between HSL and MSE exists
 - An indirect path represents the effect of one variable on another, as mediated by one or more variables
- The indirect path suggests that the relationship between High School Experience (HSL) and Math Self-Efficacy is mediated by College Experience (CC)
 - More formally, the mediational relationship is hypothesized by the path model, a formal test of hypothesis is needed to establish College Experience as a mediator of High School Experience and Math Self-Efficacy
- A number of other indirect paths exist in the model



- The indirect effect of High School Experience on Math Self-Efficacy is the combination of two path coefficients:
 - The path between High School (HSL) and College (CC) Experience: $\beta_{HSL}^{CC} = 0.662$
 - The path between College Experience (CC) and Math Self-Efficacy (MSE): $\beta_{CC}^{MSE} = 0.393$
- The **indirect effect** of HSL on MSE is the product of these two terms:
$$\beta_{HSL}^{CC} \beta_{CC}^{MSE} = 0.662 * 0.393 = 0.260$$
- The indirect effect is the amount of increase in the outcome variable (MSE in this case) that comes indirectly by a one-unit increase in the initiating variable (HSL in this case)
 - As HSL increases by one unit, CC increases by 0.662 (the direct effect of HSL on CC)
 - Then, as CC increases by 0.662, MSE increases by 0.393 (the direct effect of CC on MSE)
- Indirectly, MSE increases by 0.260 (the multiplication of the two direct effects) for every one unit increase of HSL

- Finally, of concern in mediational models and general path models is the total effect of one variable on another
- The **total effect** is the sum of all direct and indirect effects
 - It represents the **total** increase in the outcome variable for a one-unit increase in the initiating variable
- In our example, the total effect of High School Experience (HSL) on Math Self-Efficacy (MSE) is the sum of the direct and indirect effects:
$$\beta_{HSL}^{MSE} + \beta_{HSL}^{CC} \beta_{CC}^{MSE} = 4.138 + 0.662 * 0.393 = 4.398$$
- This means that for every one-unit increase in HSL, the total increase in MSE is 4.398
 - The direct effect represents the increase holding CC constant, which is implausible in this model

- Of importance in the understanding of mediating variables is the test of hypothesis for the indirect effect
 - If the indirect effect (the product of the two direct effects) is significant, then the third variable is said to be a mediator
- Hypothesis tests for the indirect effect have become a hot topic in recent years
 - This test uses a bootstrap (resampling) technique to get the p-value
- In lavaan, first label parameters:

```
#indirect effect of interest:
ind_hsl_mse := b_hsl_cc*b_cc_mse
```
- Then add effects:

```
#total effect of interest:
tot_hsl_mse := b_hsl_mse + (b_hsl_cc*b_cc_mse)
```

#Model 08: Adding indirect effects to full path model -

```
model08.syntax = "
#endogenous variable equations
perf ~ hsl + msc + mse
use ~ mse
mse ~ b_hsl_mse*hsl + b_cc_mse*cc + gender
msc ~ mse + cc + hsl
cc ~ b_hsl_cc*hsl
hsl ~ gender
```

- Lavaan provides the total and indirect effects between terminating and originating variables
- If the `standardized=TRUE` command is included in the `summary()` function call, the standardized versions of these effects are also given (the increase in standard deviations)

```

               Estimate Std.err Z-value P(>|z|)   std.lv std.all
Defined parameters:
  ind_hsl_mse      0.260   0.111   2.339   0.019   0.260   0.029
  tot_hsl_mse      4.398   0.414  10.618   0.000   4.398   0.488
```

- Here, our output suggests the indirect effect is significant, so we say that CC mediates the relationship between HSL and MSE

ADDITIONAL MODELING CONSIDERATIONS IN PATH ANALYSIS

- The path analysis we just ran was meant to be an introduction to the topic and the field
 - It is much more complex than what was described
- In particular, our path analysis assumed all variables to be
 - Continuous and Multivariate Normal
 - Measured with perfect reliability
- In reality, neither of these are true
- Structural equation models (path models with latent variables) will help with variables with measurement error
- Modifications to model likelihoods or different distributional assumptions will help with the normality assumption

- You will read a lot of talk about path models indicating causality, or how path models are causal models
- It is important to note that causality can rarely, if ever, be inferred on the basis of observational data
 - Experimental designs with random assignment and manipulations of factors will help detect causality
- With observational data, about the best you can say is that IF your model fits, then causality is ONE reason
 - But realistically, you are simply describing covariances of variables in more fancy ways/parameters
- If your model does not fit, the causality is **LIKELY** not occurring
 - But still could be possible if important variables are omitted

WRAPPING UP

- In this lecture, we discussed the basics of path analysis
 - Model specification/identification
 - Model estimation
 - Model fit (necessary, but not sufficient)
 - Model modification and re-estimation
 - Final model parameter interpretation

- There is a lot to the analysis – but what is important to remember is the over-arching principal of multivariate analyses: covariance between variables is important
 - Path models imply very specific covariance structures
 - The validity of the results hinge upon accurately finding an approximation to the covariance matrix